

Camshaft — LH



Material	
Item	Specification
Motorcraft® SAE 5W-20 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)	WSS-M2C945-A

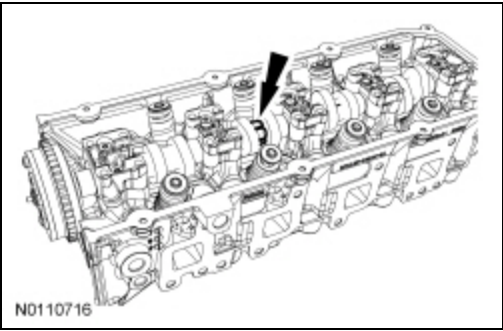
Removal

1. Remove the timing drive components. For additional information, refer to Timing Drive Components in this section.
2. **NOTICE: Damage to the camshaft phaser and sprocket assembly will occur if mishandled or used as a lifting or leveraging device.**

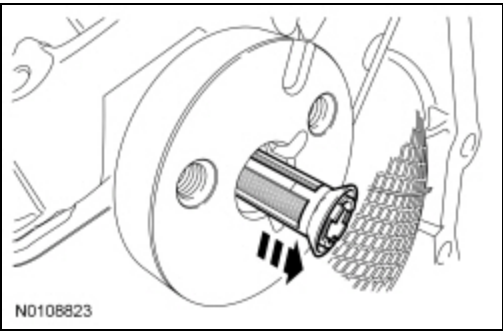
NOTICE: Only use hand tools to remove the camshaft phaser and sprocket assembly or damage may occur to the camshaft or camshaft phaser and sprocket.

Using a 26 mm (1.023 in) wrench on the flats of the camshaft to secure the camshaft, remove the 3 LH camshaft phaser and sprocket bolts and the camshaft phaser and sprocket.

- Discard the 3 camshaft phaser and sprocket bolts.



3. Remove and discard the Variable Camshaft Timing (VCT) system oil filter.



4. **NOTICE: The camshaft bearing caps must be installed in their original locations. Record camshaft bearing cap locations. Failure to follow these instructions may result in engine damage.**

Remove the 10 bolts and remove the 5 camshaft bearing caps.

5. Remove the camshaft.

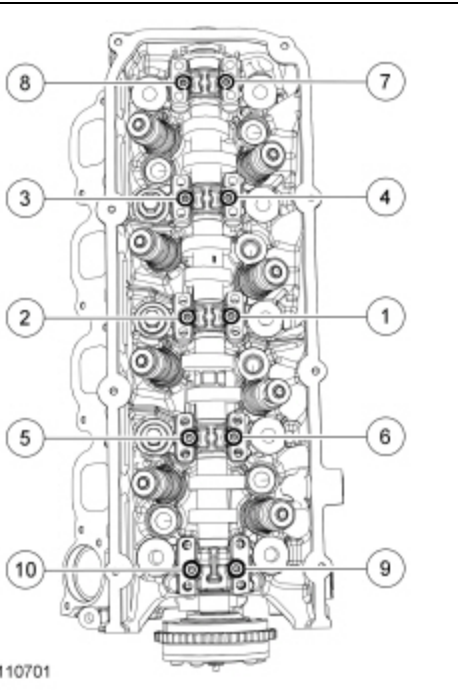
- Inspect the camshaft. For additional information, refer to Section 303-00.

Installation

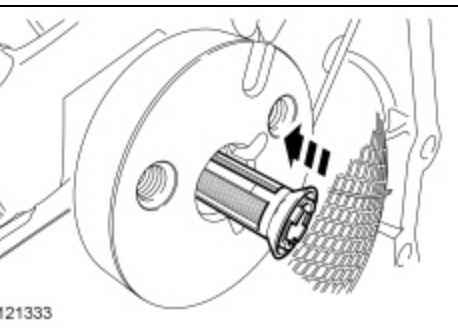
1. Lubricate the camshaft and camshaft journals with clean engine oil and install the camshaft.

2. Position the 5 camshaft bearing caps in their original locations and install the 10 bolts in the sequence shown, in 2 stages.

- Stage 1: Tighten to 6 Nm (53 lb-in).
- Stage 2: Tighten an additional 45 degrees.



3. Install a new VCT system oil filter with the open end facing the front of the engine.



4. **NOTICE: Damage to the camshaft phaser and sprocket assembly will occur if mishandled or used as a lifting or leveraging device.**

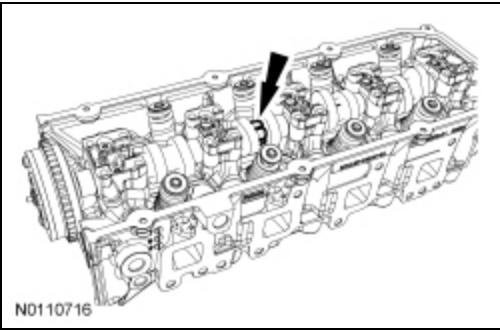
Position the camshaft phaser and sprocket with the alignment pin in the alignment slot of the camshaft and install 3 new camshaft phaser and sprocket bolts finger-tight.

5. **NOTICE: Damage to the camshaft phaser and sprocket assembly will occur if mishandled or used as a lifting or leveraging device.**

NOTICE: Only use hand tools to install the camshaft phaser and sprocket bolt or damage may occur to the camshaft or camshaft phaser and sprocket.

Using a 26 mm (1.023 in) wrench on the flats of the camshaft to secure the camshaft, tighten the 3 new LH camshaft phaser and sprocket bolts in 2 stages.

- Stage 1: Tighten to 9 Nm (80 lb-in).
- Stage 2: Tighten an additional 90 degrees.



6. Install the timing drive components. For additional information, refer to Timing Drive Components in this section.